

1. Do the following problems from the book: Challenges 20.1, 20.2, 20.4, 20.7

1.1. Challenge 20.1

- a) I create a function called 'volterra' to compute the population according to the Volterra model.

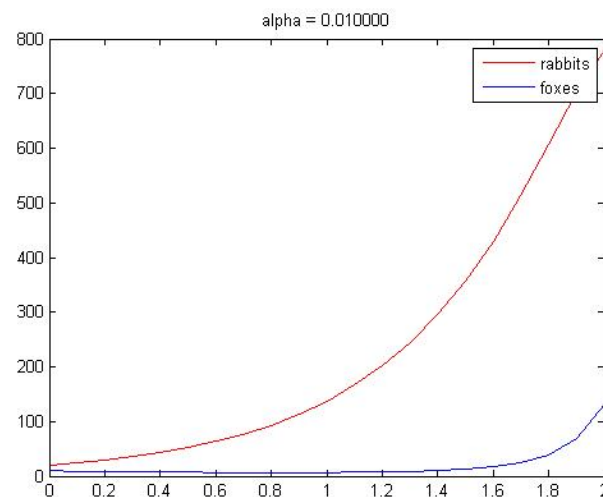
```
function f = volterra(t,y)
% compute the population according to the Volterra model.
global alpha
f(1,1) = (2 * y(1)) - (alpha*y(1)*y(2));
f(2,1) = -y(2) + (alpha*y(1)*y(2));
end
```

Then, I use 'ode45' to explore the Volterra model as follows.

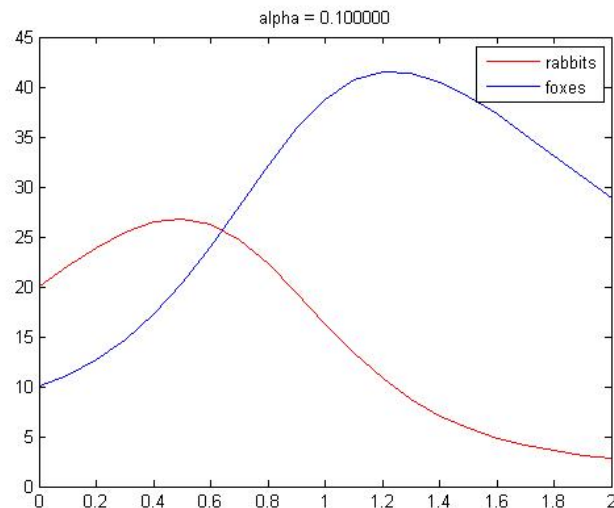
```
global alpha

for i=2:-1:0,
    alpha = 10^(-i)
    [t,y] = ode45(@volterra,[0:.1:2], [20,10]);
    plot(t,y(:,1),'r',t,y(:,2),'b');
    legend('rabbits','foxes')
    title(sprintf('alpha = %f',alpha));
    drawnow
    disp(sprintf('Calculation for alpha=%f
completed.',alpha))
    disp('Pausing. Press any key to continue.')
    pause
end
```

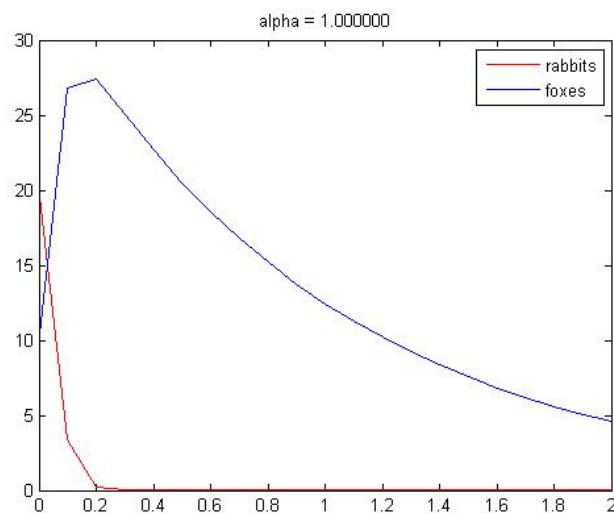
The following tree plots correspond to the tree values of alphas.



The rabbits reproduce so fast that the foxes have plenty of foods.



The rabbits are eaten by the foxes and its population drops to be less than foxes. Then fox population starts to drop.



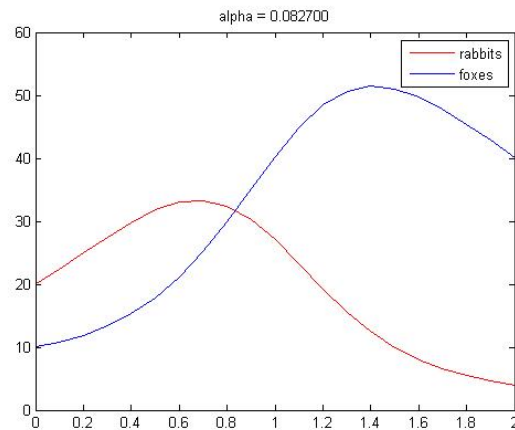
The rabbits are eaten by the foxes and then the foxes die from starvation.

b) I created a function func_14 that call ode45 which solves Volterra equation.

```
function g = func_14(a)
global alpha
alpha = a;
[t,y] = ode45(@volterra,[0:.1:2], [20,10]);
g = y(end,1) - 4;
end
```

Then, I call fzero as follows and get the alpha = 0.0827. The resulting plot indicates the final rabbit population of 4.

```
c = fzero(@func_14,2);
```



c) I create the count_zero function as follows.

```
function [VALUE, ISTERMINAL, DIRECTION] = count_zeros(T,Y)
% say we only want zeros on Y(1) - 4
VALUE = Y(1) - 4;
ISTERMINAL = 1;
DIRECTION = 0;
```

Then, I add the 'Events' option to ode45.

```
function g = func_15(a)
global alpha
alpha = a;
options = odeset('Events',@count_zeros);
[t,y,ti,yi] = ode45(@volterra,[0:.1:2], [20,10],options);
g = yi;
end
```

After that, I use fzero to search through alpha values.

```
c = fzero(@func_15,2);
```

The resulting alpha is 0.0827 which is the same as before.

1.2. Challenge 20.2

All eigenvalues should be negative. Hence, the equation will be stable when ' $t > 2$ '. This is from solving $4 - t^2 < 0$ and $-t < 0$.

1.3. Challenge 20.4

$$y'(t) = 1, y(0) = 0, h = 0.1$$

$$y_{i+1} = y_i + h * f(x_i, y_i)$$

$$y_{i+1} = y_i + 0.1$$

$y(0.1)$	0.1
$y(0.2)$	0.2
$y(0.3)$	0.3
$y(0.4)$	0.4

$y(0.7)$	0.7
$y(0.8)$	0.8
$y(0.9)$	0.9
$y(1)$	1.0

1.4. Challenge 20.7

$$y'(t) = y^2(t) - 5t$$

$$y(0) = 1$$

$$h = 0.1$$

Apply Euler's method,

$$y_{i+1} = y_i + h * f(x_i, y_i)$$

$$y(0.1) = 1 + 0.1(1*1) = 1.1$$

Evaluate,

$$f_{n+1} = f(t_{n+1}, y_{n+1}) = 1.1^2 - 0.5 = 0.71$$

Apply Backward Euler's method,

$$y_{i+1} = y_i + h * f(x_{i+1}, y_{i+1})$$

$$y(0.1) = 1 + 0.1 * 0.71 = 1.071$$

Evaluate,

$$f_{n+1} = f(t_{n+1}, y_{n+1}) = 1.071^2 - 0.5 = 0.647 \#$$

- Look up the stiffness flame example from pages 19-21 of the Moler [book](#). Reproduce the results in Figures 7.4 and 7.5. Comment on the behavior.

Figure 7.4 was reproduced by 'ode45' with the following code.

```
F = @(t,y) y^2 - y^3;
opts = odeset('RelTol',1.e-4);
delta = 0.00001;
ode45(F,[0 2/delta],delta,opts);
```

The reproduced figures are as follows. The solutions in the steady state are varying slowly but nearby solutions are varying rapidly like in the definition of stiffness.

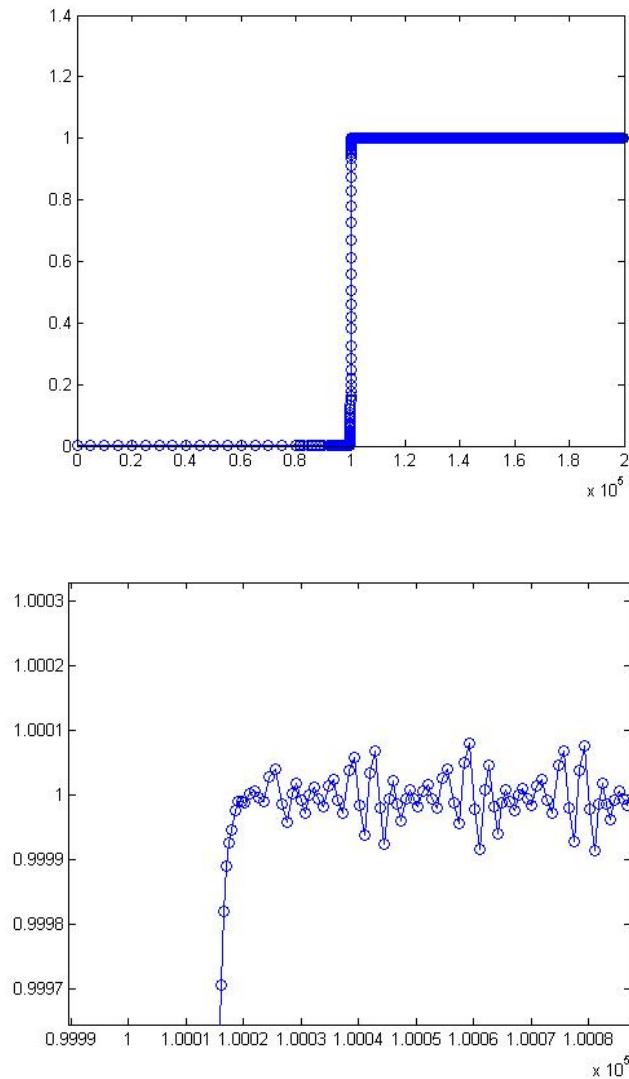
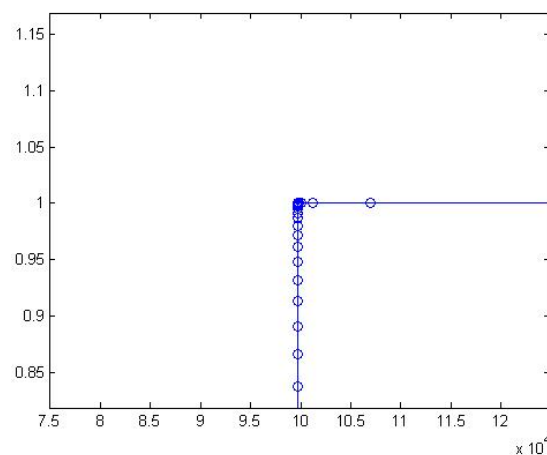
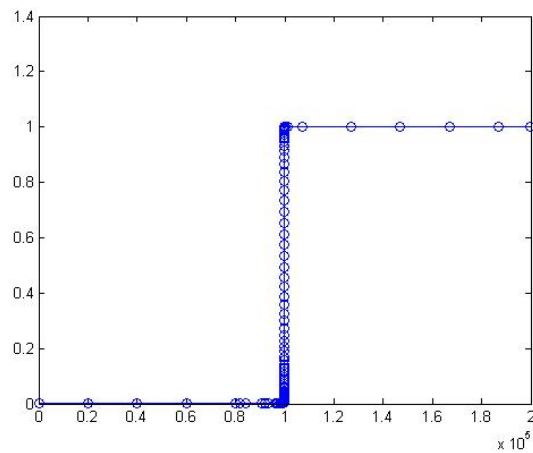


Figure 7.5 was reproduced by 'ode23s' where 's' is for stiff with the following code.

```
F = @(t,y) y^2 - y^3;
opts = odeset('RelTol',1.e-4);
delta = 0.00001;
ode23s(F,[0 2/delta],delta,opts);
```

Since 'ode23s' is a stiff solver, the solution are not varying like in 'ode45'. The reproduced figures are as follows.



3. Let $y' = y^2 - 5t$ and let $y(0) = 1$. Is the problem stable or unstable at $t = 0$? Apply three steps of the Euler and the modified Euler method, with a step size of 0.1 for the above problem, and compute approximations for $y(0.1)$, $y(0.2)$ and $y(0.3)$

The problem is unstable at $t = 0$ since the Jacobi $J(t, y) = 2$.

Similar to challenge 20.7 in question 1;

For $y(0.1)$;
 $y'(t) = y^2(t) - 5t$
 $y(0) = 1$
 $h = 0.1$
 Apply Euler's method,
 $y_{i+1} = y_i + h * f(x_i, y_i)$
 $y(0.1) = 1 + 0.1(1^2 - 0.5) = 1.05$ #
 Apply Backward Euler's method,
 $y_{i+1} = y_i + h * f(x_{i+1}, y_{i+1})$
 $y(0.1) = 1 + 0.1 * (1.1^2 - 0.5) = 1.071$ #

For $y(0.2)$;
 $h = 0.1$

$t = 0.1$

Apply Euler's method,

$$y_{i+1} = y_i + h * f(x_i, y_i)$$

$$y(0.2) = 1.1 + 0.1(1.1*1.1 - 5*0.1) = 1.171 \#$$

Apply Backward Euler's method,

$$y_{i+1} = y_i + h * f(x_{i+1}, y_{i+1})$$

$$y(0.2) = 1.071 + 0.1 * (1.171^2 - 5*0.2) = 1.081 \#$$

For $y(0.3)$;

$h = 0.1$

$t = 0.2$

Apply Euler's method,

$$y_{i+1} = y_i + h * f(x_i, y_i)$$

$$y(0.3) = 1.171 + 0.1(1.171*1.171 - 5*0.2) = 1.2081\#$$

Apply Backward Euler's method,

$$y_{i+1} = y_i + h * f(x_{i+1}, y_{i+1})$$

$$y(0.3) = 1.081 + 0.1 * (1.2081*1.2081 - 5*0.3) = 1.077\#$$

This shows the unstability of Euler's method and the stability of modified Euler's method.